DS-4021: Analytics II: Machine Learning

Syllabus

School of Data Science, University of Virginia

Subject Area and Catalog Number: Data Science, DS-4021

Year, Term: 2026, Fall

Level: Undergraduate

Credit Type: Grade (A–F)

Meeting Days: Tuesday and Thursdays

Meeting Time: 9:30–10:45 (Section 1) and 11:00–12:15 (Section 2)

Place: Data Science Building, Room TBD

Instructor: Javier Rasero (javier.rasero@uv.es)
Office Hours: By appointment
Location: SDS 335 or Zoom (https://virginia.zoom.us/j/4379931371)

Teaching Assistants:
Section 1: TBD (alguien@virginia.edu)
Section 2: TBD (otroalguien@virginia.edu)

About the course

This course continues where Analytics I (DS-3021) left off, moving into more advanced and powerful approaches, particularly neural networks, kernel-based algorithms, and ensemble learning. You will gain a deeper understanding of core ML concepts while comparing methods and discussing how performance, complexity, and predictive power evolve as models become more sophisticated.

The course emphasizes understanding over blind implementation. More than often, you will implement models from scratch rather than relying solely on libraries. You will be challenged to evaluate their assumptions, strengths, and weaknesses. Self-learning and group work are central components, developed through a mix of labs, team-based projects, combining theory, application, discussion and reflection. The course is structured to support students from diverse backgrounds, with ample opportunities for collaboration, TA support, and office hours.

Ultimately, Analytics II seeks to provide a first experience as a real-life practicing data scientist, where advanced models and critical thinking are typically required. It also aims to expand your toolkit with new methods and encourages you to share discoveries with your peers as you improve your ability to generate meaningful insights from data.

What you will learn

These are the main learning objectives:

- Deepen their understanding of machine learning by exploring more advanced and state-of-the-art methods, including support vector machines, ensemble techniques, and neural networks.
- Understand the importance of structured workflows in solving data science problems.
- Gain first exposure to implementing machine learning models from scratch using PyTorch and applying them to real-world datasets.
- Apply machine learning models real-world datasets using the gold-standard library for machine learning, (scikit-learn.
- Critically evaluate and compare the performance of different models, considering trade-offs between accuracy, interpretability, and computational cost.
- Develop and document reproducible data science workflows, including code annotation, version control (GitHub), and final deliverables.
- Engage in collaborative, project-based learning by working effectively within a group to complete lab assignments and a final project.
- Demonstrate self-directed learning and adaptability by tackling tasks beyond the core material and seeking solutions independently.
- Communicate technical findings clearly and effectively, both in written reports and oral presentations.

- Reflect on individual contributions, strengths, and areas for improvement, incorporating peer and self-critique into a growth-oriented learning process.

How you will learn

Each week, you will be expected to review short video lectures and complete assigned readings before coming to class. The videos serve as a brief introduction to the main concepts, which will be expanded upon during the first class of the week through a hands-on Jupyter Notebook walkthrough.

Since Data Science is inherently collaborative, nearly every week will include a group-based lab session. These labs are designed not only to help you practice what you have learned that week, but also to push you out of your comfort zone by encouraging you to learn something new. Machine Learning requires a strong element of self-directed learning, and these sessions will help you build that skill. Both the TA and I will be available during and after class to provide any support you need.

Groups will be formed at the beginning of the semester by balancing skill levels, and you will work with the same group throughout the course.

Evaluation Criteria

- **Labs (50%):** Throughout the course, and in most weeks, we will have eight in-class graded labs. These are designed to help you either practice the skills presented in class or explore new material related to the week's theme. These are **group assignments**, and you are expected to work closely with your group mates to divide and complete the different tasks. The main deliverable will be a completed Jupyter Notebook, submitted by the group, which will determine the grade for the assignment. Additionally, each member of the team will submit an individual report describing the specific tasks they contributed to. Each member must submit this individual report separately in order to receive a grade. The deadline for submission of all required materials is typically **one week** after their publication on Canvas.

- **Final Project (30%):** The course will culminate in a final project involving work with a dataset of your choice. This is a combined **group** and **individual** assignment, and you will work with the same group with which you submitted the lab assignments. As a group, you will submit a final GitHub repository, including all well-annotated code and a final report summarizing your findings. Individually, each member of the team will also submit a GitHub repository containing the code they contributed to different parts of the project, as well as an individual report describing the specific tasks they contributed to. Each member must submit their individual GitHub repository separately in order to receive a grade. **The final project grade will be the average of the group component and the individual component**. This is an open-ended project designed to allow each group to explore a topic of interest from the semester in greater depth.

- **Final Exam (20%):** This will be an **individual** written exam covering material from throughout the course. The exam will take place in class on the date stipulated by the University for this course.

Materials

A. Python Machine Learning with PyTorch and Scikit-Learn, 4th Edition. Freely available on O'Reilly with your UVA account.

B. Introduction to statistical learning. Free PDF.

C. The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Free PDF.

D. Pattern Recognition and Machine Learning. Free PDF.

Tech Stack

- JupyterLab/Jupyter-notebook
- UVA Open OnDemand
- Canvas

A few Policies that will Govern the Class

Grading Policies: Courses carrying a Data Science subject area use the following grading system: A, A-; B+, B, B-; C+, C, C-; D+, D, D-; F. The symbol W is used when a student officially drops a course before its completion or if the student withdraws from an academic program of the University.

Grading Scale:

- 93-100 A
- 90-92 A-
- 87-89 B+
- 83-86 B
- 80-82 B-
- 77-79 C+
- 73-76 C
- 70-72 C-
- < 70 F

Collaboration and Conflict Resolution: Should any conflict arise within your group—for example, uneven workload distribution or communication breakdowns—please first make an effort to address the issue respectfully among your group members. If the problem persists or becomes difficult to resolve, group members should reach out to the instructor or TA as early as possible. Addressing concerns early allows us to work together toward a constructive solution before the issue impacts your work or group dynamic.

Late work: Late submissions will NOT be accepted except in cases of emergency, such as illness or family matters. If you anticipate a conflict in advance—for example, due to athletics or another university activity—please reach out ahead of time so we can arrange a suitable alternative for submission.

Use of Large Language Model tools: The use of Large Language Models (e.g., ChatGPT) is acceptable and even encouraged for the coding aspects of the lab assignments and final project, although with caveats. You need to remember that they may make mistakes, so any code generated by this kind of tools should be carefully checked. On the other hand, using LLM in the writing parts of the assignments and final project is discouraged, as it may hinder your own learning. Finally, their use is strictly prohibited during and for quizzes.

Religious Accomodations: It is the University's long-standing policy and practice to reasonably accommodate students so that they do not experience adverse academic consequences when sincerely held religious beliefs or observances conflict with academic requirements. Students who wish to request academic accommodations for a religious observance should submit their request to the instructor by email as far in advance as possible. If you have questions or concerns about your request, you can contact the University's Office for Equal Opportunity and Civil Rights (EOCR) at UVAEOCR@virginia.edu or (434) 924-3200. Accommodations do not relieve you of the responsibility for the completion of any part of the coursework you miss as the result of a religious observance.

University of Virginia Honor System: By enrolling in this course, all students are expected to uphold the University of Virginia Honor System (https://honor.virginia.edu/course/honor-code). This includes abiding by the Honor Code in all aspects of the course.

Special Needs: The University of Virginia accommodates students with disabilities. Any SCPS student with a disability who needs accommodation (e.g., in arrangements for seating, extended time for examinations, or note-taking, etc.), should contact the Student Disability Access Center (SDAC) and provide them with appropriate medical or psychological documentation of his/her condition. Once accommodations are approved, just follow up with me concerning any logistics and implementation of accommodations. Please try to make accommodations for test-taking at least 14 business days in advance of the date of the test(s). Students with disabilities are encouraged to contact the SDAC: 434-243-5180/Voice, 434-465-6579/Video Phone, 434-243-5188/Fax. Further policies and statements are available at www.virginia.edu/studenthealth/sdac/sdac.html.

Class Schedule (subject to change)

Week	Tuesday Class	Thursday Class	Suggested Readings
Module 1: ML1 Bridge			
25-27 August	Course Overview. Skill Assessment (for Group Formation)	Feature wrangling: transformations, missingness, scaling, selection.	
1-3 September	Mastering model evaluation and optimization	Introduction to PyTorch	
Module 2: Dealing with High Dimensional Data			
8-10 September	Regularized Regression through PyTorch and Stochastic Gradient Descent	Principal Component Analysis as a preprocessing step. Introduction to Pipelines	
15-17 September	**Lab**: Partial Least Squares Regression (PLS)	Support Vector Machine I: The Linear Case	
Module 3: Exploiting Non-Linear Relationships in Data			
22-24 September	Support Vector Machine II: The Non-Linear case through the Kernel Trick	**Lab**: Predicting Seismic Building Response with Support Vector Regrssion	A. Chapter 3: Maximum margin classification with support vector machines
29 September-1 Octobe	Kernels are EVERYWHERE	**Lab**: Non-linear dimensionality reductions: t-SNE, and UMAP	A. Chapter 3: Solving nonlinear problems using a kernel SVM
Module 4: Neural Networks			
6-8 October	Reading Day (No Class)	Neural Networks I: Introduction & Multi-Layer Perceptrons (MLPs)	

13-15 October	**Lab**: Manual implementation of MLPs for Classification and Regression using toy data	Neural Networks II: Deeper into model building, training and opmitization.	A. Chapter 11: Modeling complex functions with artificial neural networks
20-22 October	**Lab**: Application of Neural Networks to Real Data	Neural Networks III: Improving generalization	A. Chapter 13: Making model building more flexible with nn.Module
27-29 October	**Lab**: Implementing an autoencoder for non-linear dimensionality reduction	Bagging: Bootstrap Aggregating and Random Forest	
Module 5: Ensemble Methods & Meta-Learning			
3-5 November	Election day (No class)	Boosting: Adaptive Boosting and Gradient Boosting	
10-12 November	**Lab**: Classifying Internet of Things Device Types with Bagging and Boosting	Clustering I: k-Means and Gaussian Mixture Models	
Module 6: Tackling Population Heterogeneity			
17-19 November	Clustering II: Hierarchical Clustering and DBSCAN	Clustering III: Validation	
26-28 November	**Lab**: clustering	Thanksgiving (No class)	
1-3 December	Final Project Time		